

Problem Set 1

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1 Propositional logic

Problem 1: Express all of the following statements using propositional logic. Declare your atomic propositions (e.g. W = “I am the Walrus”).

1. Ted is dead or Ted is not dead
2. If Ted is dead and Ted is not dead then Ted is a plant.
3. If Bob is here implies that Bob is lost, then Bob is lost.
4. Ted is a ghost if and only if Ted is dead. Also, Ted is a ghost and Ted is not dead.

2 Truth Tables

Problem 2: Write out truth tables for each of the propositions above. Label them tautologies (always true), contradictions (always false) or satisfiable (sometimes true).

Problem 3: Write out the truth tables for the following propositions:

1. $\neg(P \wedge Q) \iff \neg P \vee \neg Q$
2. $(P \wedge Q) \rightarrow R \iff P \rightarrow (Q \rightarrow R)$